SECTION A

1. (a) A beam ACB is subject to a point load P at hinge C as shown in Figure 1(a).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tension** side. **(5 marks)**

   (b) A frame is subject to a uniformly distributed load q applied along BC, as illustrated in Figure 1(b). A is a roller support and B is a hinge support.

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tension** side. **(5 marks)**

2. As case 1 shown in Figure 2(a), two beams are piled up without bonding in the interface, see Figure 2(b) for the cross sections, whereas for case 2 in Figure 2(c), they are pinned together with the cross sections shown in Figure 2(d). For each case, the two beams are fixed to the wall at point A, and a concentrated load F is applied at the free end B. The upper beam and bottom beam have the same Young's Modulus E but different thicknesses, h and $3h$ respectively. Young's Modulus E= 200 GPa, width of cross section b= 6 cm, thickness of upper beam h= 2 cm, beam span L=1 m.

   (a) If F= 10.08kN, calculate the deflections $w_{B,1}$ at point B for case 1. **(4 marks)**

   (b) Using the same load F in Question (b), calculate the deflections $w_{B,2}$ at point B for case 2. **(4 marks)**

   (c) If the allowable stress for the material is $[\sigma]$= 250MPa, find the maximum allowable load $F_{1,max}$ and determine the failure surface for case 1. Only consider strength condition and assume the tensile strength equals to the compressive strength. **(6 marks)**

   (d) Using the same allowable stress $[\sigma]$ in Question (d), find the maximum allowable load $F_{2,max}$ and determine the failure surface for case 2. Only consider strength condition and assume the tensile strength equals to the compressive strength. **(6 marks)**

3. (a) A simply supported beam AB is subjected to point load F at the middle point C as shown in Figure 3(a) (Case A), where w is the deflection at point C. Flexural rigidity EI is a constant. Express F in terms of w, EI and L. **(4 marks)**
   Hint: For elasticity, load is proportional to deformation.

   (b) If an linear spring with stiffness k is added at point C as shown in Figure 3(b) (Case B), express F in terms of w, EI, L and k. **(4 marks)**
   Hint: According to the Hooke's Law for linear spring, restoring force F_s is proportional to displacement $\triangle$, satisfying $F_s=k\triangle$.

(c) Beam AB, with a constant flexural rigidity *EI*, is connected to a rod by pin C as shown in Figure 3(c) (Case C), where *EA* is the axial stiffness of rod CD, and *l* is the length of rod CD. Express *F* in terms of *w*, *EI*, *EA* and *L*. Rod CD will not buckle. **(4 marks)**

(d) Two identical beams AB and EF, with the same constant flexural rigidity *EI*, is pin-connected with a rod CD as shown in Figure 3(d) (Case D). If *EA* is infinity, express *F* in terms of *w*, *EI* and *L*. Rod CD will not buckle. **(4 marks)**

(e) Considering the same scenario with Question (d) except that *EA* is a finite value as shown in Figure 3(e) (Case E), express *F* in terms of *w*, *EI*, *EA* and *L*. Rod CD will not buckle. **(4 marks)**

FIGURES

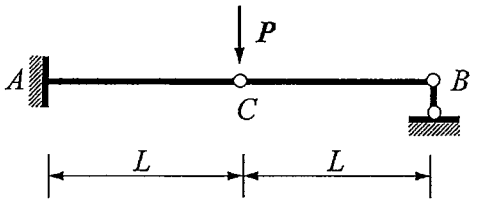


Figure 1(a) Beam

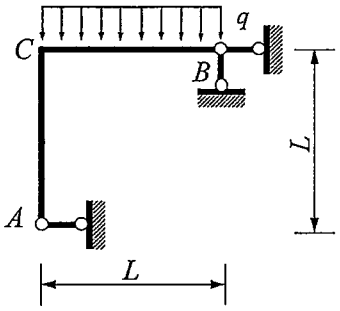


Figure 1(b) Frame

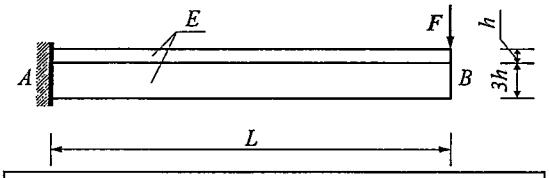


Figure 2(a) Case 1

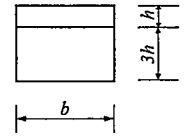


Figure 2(b) Cross section of case 1

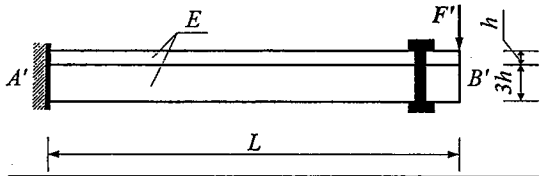


Figure 2(c) Case 2

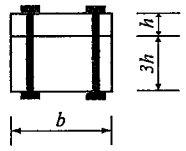


Figure 2(d) Cross section of case 2

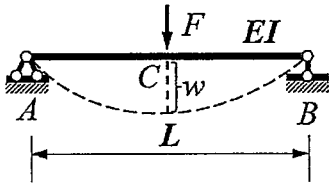


Figure 3(a) Case A

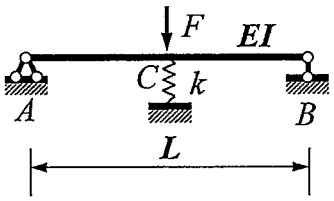


Figure 3(b) Case B

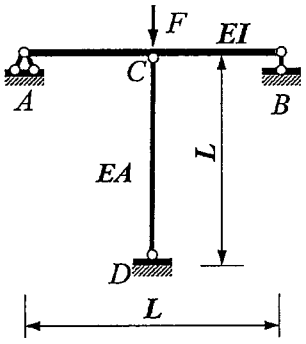


Figure 3(c) Case C

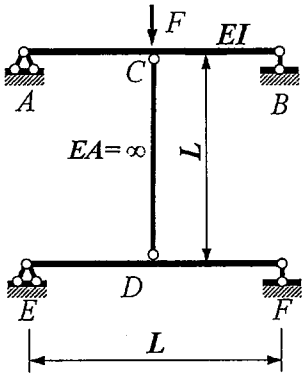


Figure 3(d) Case D

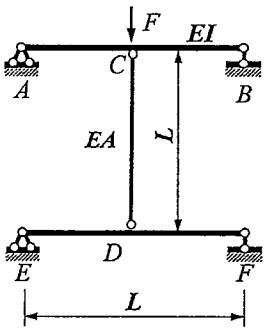


Figure 3(e) Case E

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